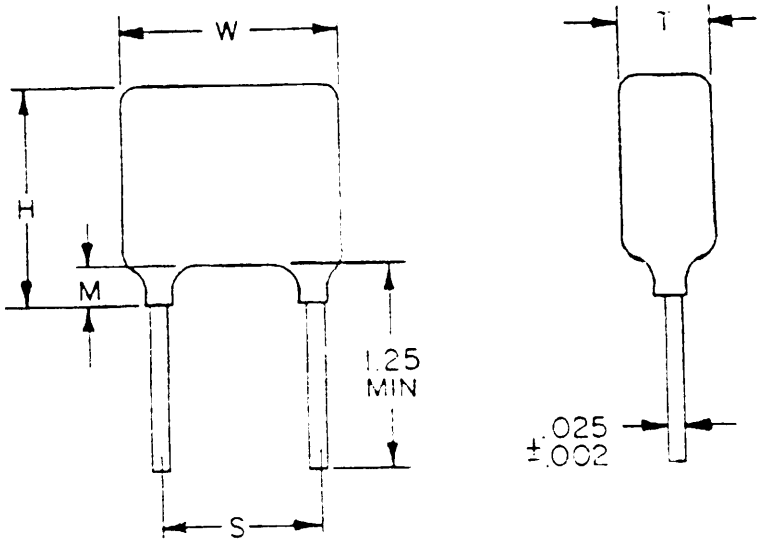
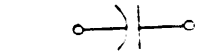
MILITARY SPECIFICATION SHEET

Capacitors, Ceramic, Multilayer, High Voltage, 2000 V DC, Established Reliability, Style HVR02

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49467.





CIRCUIT DIAGRAM

Case code	Dimensions				
	W max	H max	T max	M max	S ±.030
A	.250	.220	.200	.045	.170
B	.320	.280	.250	.045	.220
C	.370	.300	.250	.045	.275
D	.420	.400	.250	.060	.325
E	.570	.500	.270	.060	.475
F	.670	.600	.270	.060	.575
G	.770	.720	.270	.060	.675

Inches	mm	Inches	mm
.002	.05	.370	9.40
.025	.64	.375	9.52
.030	.76	.400	10.16
.045	1.14	.470	11.94
.060	1.52	.475	12.06
.170	4.32	.500	12.70
.200	5.08	.570	14.48
.220	5.59	.600	15.24
.250	6.35	.670	17.02
.270	6.86	.675	17.14
.275	6.98	.720	18.29
.280	7.11	.770	19.56
.300	7.62	1.250	31.75
.320	8.13		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Capacitor style HVR02.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Conformally coated, molded, or encapsulated.

Lead material: In accordance with MIL-STD-1276.

Capacitance value: See table I.

Capacitance tolerance: J = ±5%, K = ±10%, M = ±20%. See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: 2000 V dc.

Failure rate level: M (1.0%), P (0.1%), R (.01%), or S (.001%).

Dielectric types: BP, BR, and BZ in accordance with MIL-C-49467.

Temperature coefficient: 0 ±30 ppm/°C (BP), ±15% (BR, BZ).

Voltage-temperature limits: In accordance with MIL-C-49467.

Symbol	Capacitance change with reference to +25°C	
	Steps A through D of voltage-temperature limit cycle table of MIL-C-49467. Bias = 0 volt.	Steps E through G of voltage-temperature limit cycle table of MIL-C-49467.
BP	0 ±30 ppm/°C	100% rated voltage: 0 ±30 ppm/°C
BR	±15%	100% rated voltage: +15, -40% 40% rated voltage: +15, -25% 60% rated voltage: +15, -40%
BZ	±15%	

Dissipation factor (25°C): 2.5% maximum (BR and BZ characteristics) or 0.1% maximum (BP characteristic), measured under the same conditions as capacitance.

Resistance to soldering heat: ΔC: -1% to +2% for BP characteristic.
-1% to +6% for BR and BZ characteristics.

Marking: In accordance with MIL-C-49467.

APPLICATION NOTES:

Additional encapsulation is necessary in applications where the possibility of a voltage breakdown between the leads of the capacitor, or the capacitor to another potential, could occur.

Heat sinks on each lead or adequate preheating is required when these capacitors are installed in or removed from circuits by soldering iron.

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TABLE I. Capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467P02100__	10 pF	J, K	BP	A
M49467P02120__	12	J, K	BP	A
M49467P02150__	15	J, K	BP	A
M49467P02180__	18	J, K	BP	A
M49467P02220__	22	J, K	BP	A
M49467P02270__	27	J, K	BP	A
M49467P02330__	33	J, K	BP	A
M49467P02390__	39	J, K	BP	A
M49467P02470__	47	J, K	BP	A
M49467P02560__	56	J, K	BP	A
M49467P02680__	68	J, K	BP	A
M49467P02820__	82	J, K	BP	A
M49467P02101__	100	J, K	BP	A
M49467_02101__	100	K, M	BR, BZ	A
M49467P02121__	120	J, K	BP	A
M49467_02121__	120	K, M	BR, BZ	A
M49467P02151__	150	J, K	BP	A
M49467_02151__	150	K, M	BR, BZ	A
M49467P02181__	180	J, K	BP	A
M49467_02181__	180	K, M	BR, BZ	A
M49467P02221__	220	J, K	BP	A
M49467_02221__	220	K, M	BR, BZ	A
M49467P02271__	270	J, K	BP	A
M49467_02271__	270	K, M	BR, BZ	A
M49467P02331__	330	J, K	BP	A
M49467_02331__	330	K, M	BR, BZ	A
M49467P02391__	390	J, K	BP	B
M49467_02391__	390	K, M	BR, BZ	A
M49467P02471__	470	J, K	BP	B
M49467_02471__	470	K, M	BR, BZ	A
M49467P02561__	560	J, K	BP	B
M49467_02561__	560	K, M	BR, BZ	A
M49467P02681__	680	J, K	BP	B
M49467_02681__	680	K, M	BR, BZ	A
M49467P02821__	820	J, K	BP	B
M49467_02821__	820	K, M	BR, BZ	A
M49467P02102__	1000	J, K	BP	C
M49467_02102__	1000	K, M	BR, BZ	A
M49467P02122__	1200	J, K	BP	C
M49467_02122__	1200	K, M	BR, BZ	A
M49467P02152__	1500	J, K	BP	D
M49467_02152__	1500	K, M	BR, BZ	A
M49467P02182__	1800	J, K	BP	D
M49467R02182__	1800	K, M	BR	B
M49467Z02182__	1800	K, M	BZ	A
M49467P02222__	2200	J, K	BP	D
M49467R02222__	2200	K, M	BR	B
M49467Z02222__	2200	K, M	BZ	A
M49467P02272__	2700	J, K	BP	E
M49467R02272__	2700	K, M	BR	B
M49467Z02272__	2700	K, M	BZ	A
M49467P02332__	3300	J, K	BP	E
M49467_02332__	3300	K, M	BR, BZ	E

See footnote at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467P02392__	3900 pF	J, K	BP	E
M49467 02392__	3900	K, M	BR, BZ	B
M49467P02472__	4700	J, K	BP	E
M49467R02472__	4700	K, M	BR	C
M49467Z02472__	4700	K, M	BZ	B
M49467P02562__	5600	J, K	BP	F
M49467R02562__	5600	K, M	BR	C
M49467Z02562__	5600	K, M	BZ	B
M49467P02682__	6800	J, K	BP	F
M49467R02682__	6800	K, M	BR	D
M49467Z02682__	6800	K, M	BZ	B
M49467P02822__	8200	J, K	BP	G
M49467R02822__	8200	K, M	BR	D
M49467Z02822__	8200	K, M	BZ	C
M49467P02103__	.01 uF	J, K	BP	G
M49467R02103__	.01	K, M	BR	D
M49467Z02103__	.01	K, M	BZ	C
M49467 02123__	.012	K, M	BR, BZ	D
M49467 02153__	.015	K, M	BR, BZ	D
M49467R02183__	.018	K, M	BR	E
M49467Z02183__	.018	K, M	BZ	D
M49467R02223__	.022	K, M	BR	E
M49467Z02223__	.022	K, M	BZ	D
M49467R02273__	.027	K, M	BR	E
M49467Z02273__	.027	K, M	BZ	D
M49467 02333__	.033	K, M	BR, BZ	E
M49467R02393__	.039	K, M	BR	F
M49467Z02393__	.039	K, M	BZ	E
M49467R02473__	.047	K, M	BR	F
M49467Z02473__	.047	K, M	BZ	E
M49467R02563__	.056	K, M	BR	G
M49467Z02563__	.056	K, M	BZ	E
M49467R02683__	.068	K, M	BR	G
M49467Z02683__	.068	K, M	BZ	F
M49467R02823__	.082	K, M	BR	G
M49467Z02823__	.082	K, M	BZ	F
M49467 02104__	.10	K, M	BR, BZ	G
M49467Z02124__	.12	K, M	BZ	G
M49467Z02154__	.15	K, M	BZ	G

1/ The complete PIN shall include additional letter symbols to indicate the characteristic, where applicable, the capacitance tolerance, and the failure rate level.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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